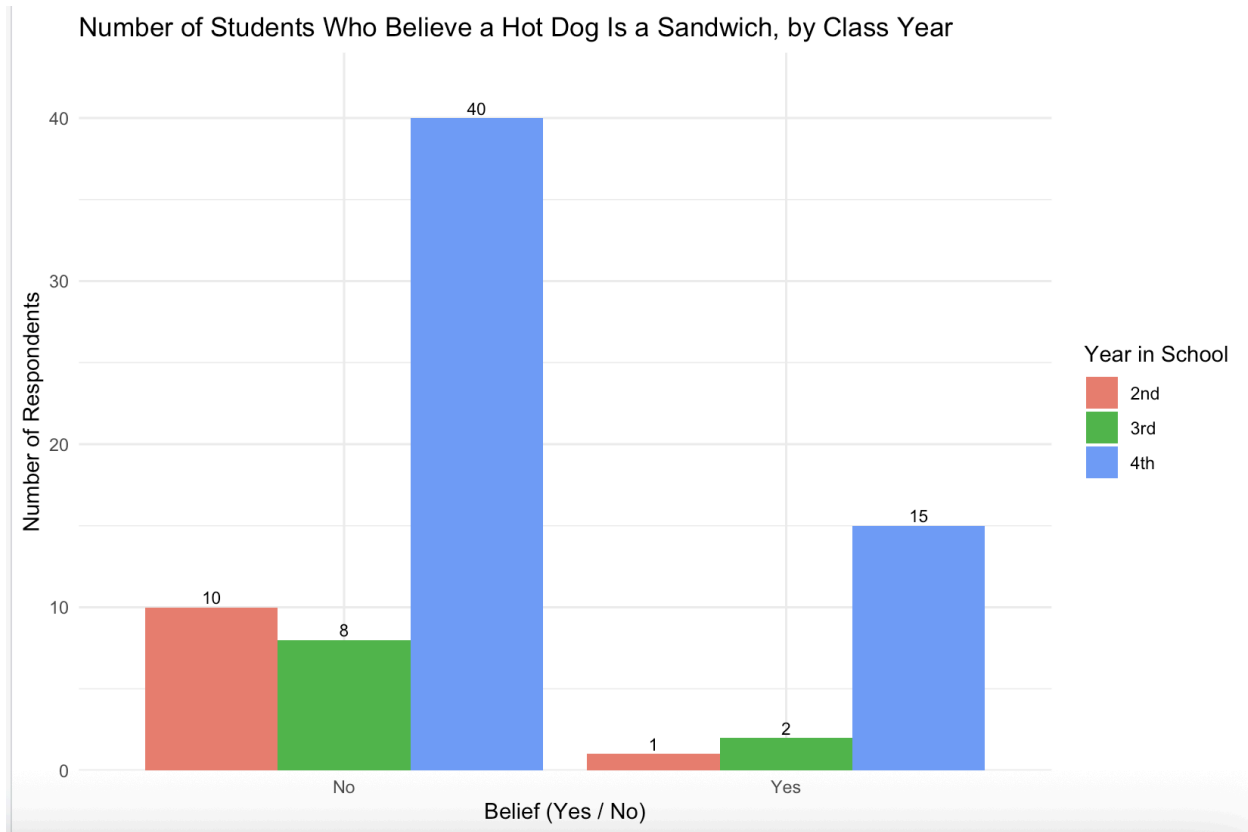


Pre-Analysis Steps: Our research question asked whether a hot dog is considered a sandwich, and we hypothesized that more than half of full-time undergraduate students at UVA would say it is not. We first looked through all 79 observations and made sure every student answered all 3 questions. Everyone answered all three questions. We then cleaned the data to omit any response that answered “No” to “Are you a full-time undergraduate UVA student.” The cleaned dataset includes 76 observations. Next, we converted the answers to a CSV file in order to use it for analysis in R.

Analysis Methods: We will create a bar chart that categorizes answers by “Yes” or “No” on the X-axis, with different colors corresponding to the year of the student. From that, we can use a chi-squared test to determine if there’s a relationship between the year of the student and their thoughts on whether or not a hot dog is a sandwich.

Evaluation of Success: Based on the bar-chart, agreement that a hot dog is a sandwich is a minority opinion across all class years. The 4th-years show both the strongest rejection (“No”) and the largest number who accept (“Yes”), which likely reflects a larger sample size rather than a shift in belief. A chi-square goodness of fit test showed that significantly more than half of full-time undergraduate UVA students believe that a Hot Dog is not a sandwich ($p < .001$) (see output below).



```
> hotdog_test
```

Chi-squared test for given probabilities

```
data: hotdog_table
```

```
X-squared = 21.053, df = 1, p-value = 4.468e-06
```

```
> hotdog_test$expected
```

```
No Yes
```

```
38 38
```

```
> hotdog_test$stdres
```

```
      No      Yes  
4.588315 -4.588315
```

```
>
```

```
>
```